无人系统设计 课程作业03

班级：电院2273
学号：522031910213
姓名：朱涵

1 作业1

1.1 参数截图

1.1.1 第一次修改

```
actorNet = [
    featureInputLayer(obsInfo.Dimension(1))
    fullyConnectedLayer(5)
    tanhLayer
    fullyConnectedLayer(actInfo.Dimension(1))
    ];
```

```
criticNet = initialize(criticNet);
summary(criticNet)
```

```
已初始化：true

可学习参数的数量：1.5k

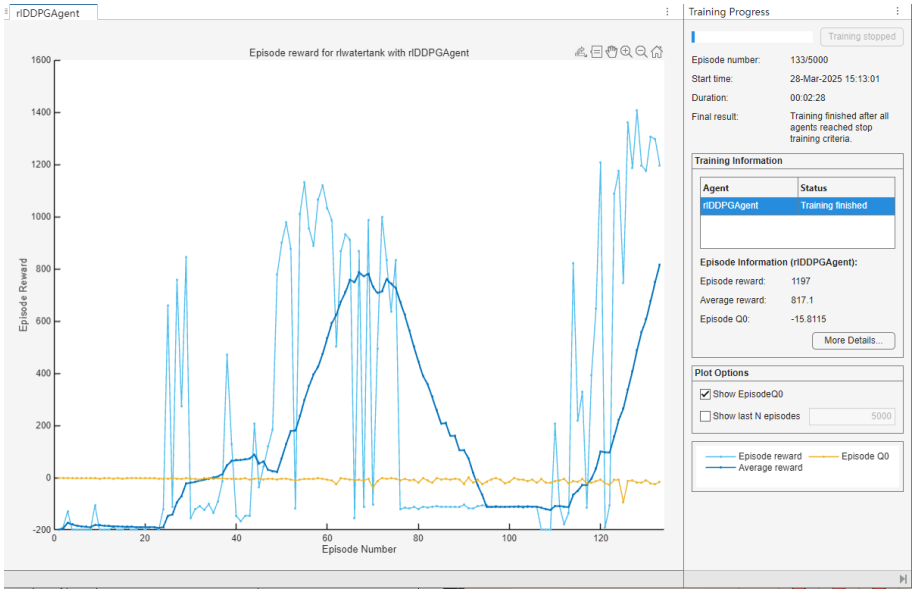
输入：
   1   'obsInLyr'   3 个特征
   2   'actInLyr'   1 个特征
```

```
actorNet = dlnetwork(actorNet);
summary(actorNet)
```

```
已初始化：true

可学习参数的数量：26

输入：
   1   'input'   3 个特征
```



	rlDDPGAgent
Status	Training finished
Episode number	133
Episode reward	1197
Episode steps	200
Total agent steps	18529
Average reward	817.1
Average steps	176.6
Episode Q0	-15.8115
Averaging window length	20
Training stopped by	AverageReward
Training stopped at	800

可以看到迭代次数增加了。这是因为网络宽度增加，每层有更多的神经元，从而增加模型参数数量。同时一个actor更宽的模型也可以更好地适应复杂的控制策略。

1.1.2 第二次修改

```
actorNet = [
    featureInputLayer(obsInfo.Dimension(1))
    fullyConnectedLayer(3)
    tanhLayer
    fullyConnectedLayer(3)
    tanhLayer
    fullyConnectedLayer(actInfo.Dimension(1))
    ];
```

```
criticNet = initialize(criticNet);
summary(criticNet)
```

已初始化：true

可学习参数的数量：1.5k

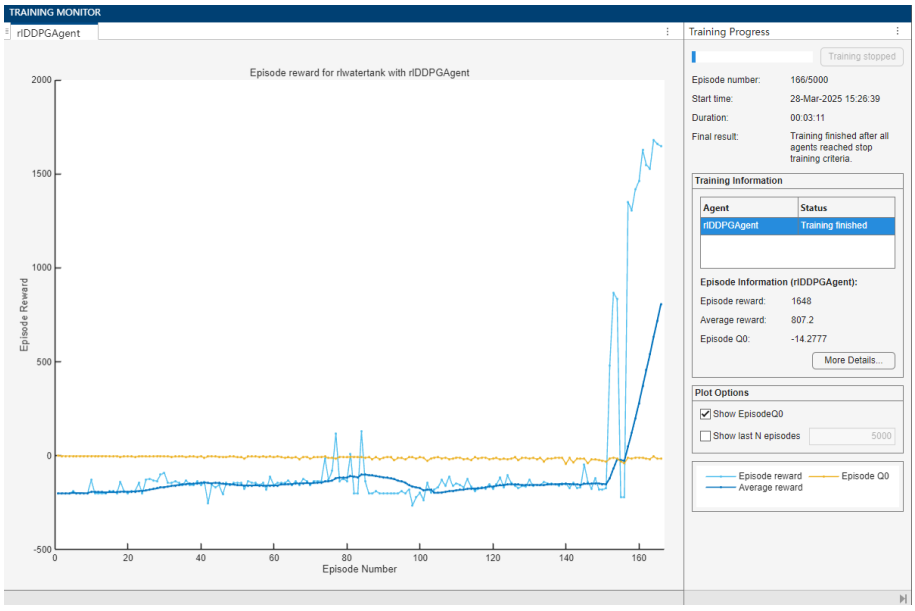
输入：
   1   'obsInLyr'   3 个特征
   2   'actInLyr'   1 个特征

```
actorNet = dlnetwork(actorNet);
summary(actorNet)
```

已初始化：true

可学习参数的数量：28

输入：
   1   'input'   3 个特征



	rlDDPGAgent
Status	Training finished
Episode number	166
Episode reward	1648
Episode steps	200
Total agent steps	23697
Average reward	807.2
Average steps	158.25
Episode Q0	-14.2777
Averaging window length	20
Training stopped by	AverageReward
Training stopped at	800

可以看到迭代次数进一步增加了。这是因为网络深度增加，模型复杂度变高。同时一个actor更深的模型也可以更好地适应复杂的控制策略。

1.1.3 第三次修改

```
% Common path
commonPath = [
    additionLayer(2,Name="add")
    reluLayer
    fullyConnectedLayer(3,Name="Hidden")
    reluLayer
    fullyConnectedLayer(1,Name="QValue")
    ];
```

```
criticNet = initialize(criticNet);
summary(criticNet)
```

```
已初始化：true

可学习参数的数量：1.6k

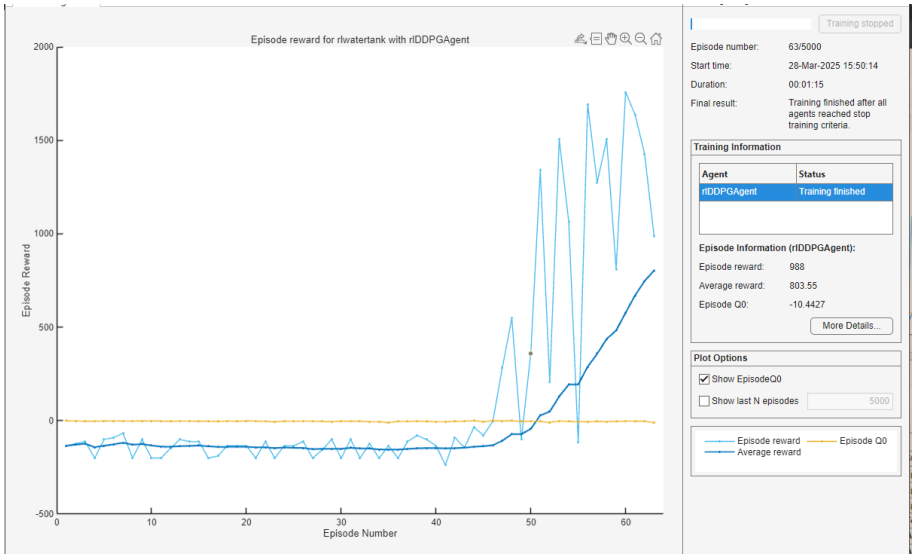
输入：
   1   'obsInLyr'   3 个特征
   2   'actInLyr'   1 个特征
```

```
actorNet = dlnetwork(actorNet);
summary(actorNet)
```

```
已初始化：true

可学习参数的数量：16

输入：
   1   'input'   3 个特征
```



	rlDDPGAgent
Status	Training finished
Episode number	63
Episode reward	988
Episode steps	200
Total agent steps	12122
Average reward	803.55
Average steps	191.9
Episode Q0	-10.4427
Averaging window length	20
Training stopped by	AverageReward
Training stopped at	800

可以看到迭代次数减少。这是因为Critic网络深度增加，模型的拟合能力变强，加快了学习进程。同时一个critic更深的模型也可以更好地适应复杂度A-C关系。

1.1.4 第四次修改

```
% Common path
commonPath = [
    additionLayer(2,Name="add")
    reluLayer
    fullyConnectedLayer(5,Name="Hidden")
    reluLayer
    fullyConnectedLayer(1,Name="QValue")
    ];
```

```
criticNet = initialize(criticNet);
summary(criticNet)
```

已初始化：true

可学习参数的数量：1.6k

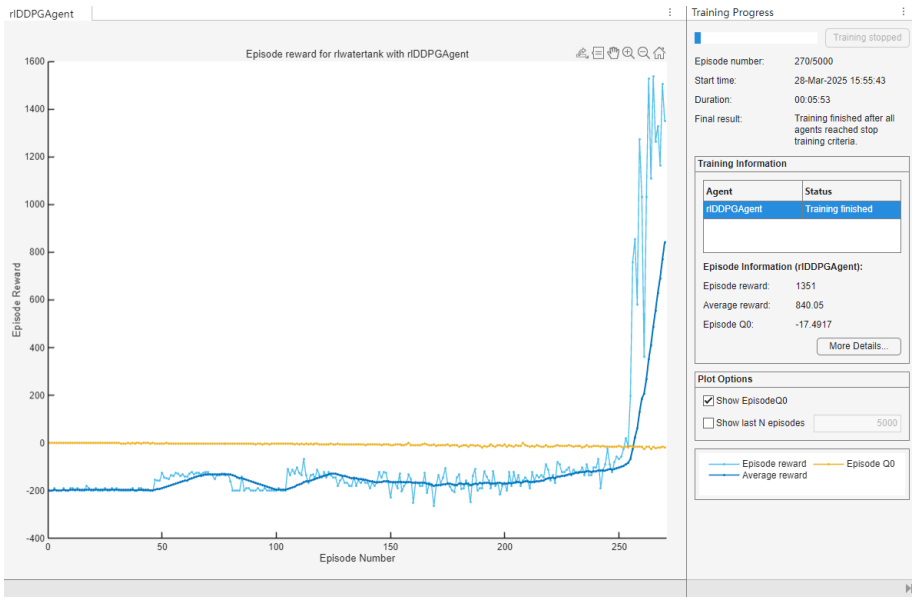
输入：
   1   'obsInLyr'   3 个特征
   2   'actInLyr'   1 个特征

```
actorNet = dlnetwork(actorNet);
summary(actorNet)
```

已初始化：true

可学习参数的数量：16

输入：
   1   'input'   3 个特征



	rlDDPGAgent
Status	Training finished
Episode number	270
Episode reward	1351
Episode steps	200
Total agent steps	43617
Average reward	840.05
Average steps	200
Episode Q0	-17.4917
Averaging window length	20
Training stopped by	AverageReward
Training stopped at	800

可以看到迭代次数大大增加了。这是因为网络宽度增加，每层有更多的神经元，从而增加模型参数数量。同时一个critic更宽的模型也可以更好地拟合复杂的Q函数，改善智能体的决策。

1.2 简表结论

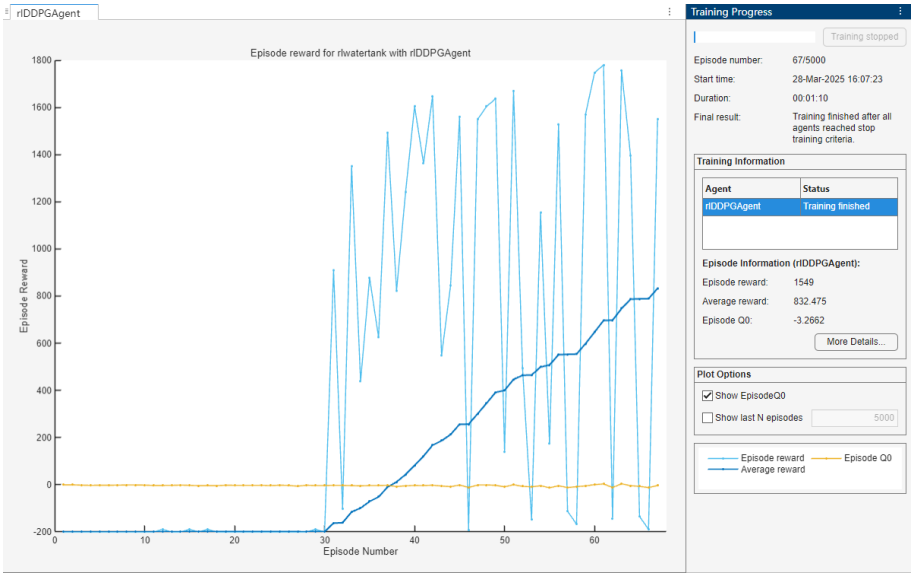
序号	Actor宽度	Actor深度	Critic宽度	Critic深度
1	5	3	1	2
2	3	4	1	2
3	3	3	3	3
4	3	3	5	3

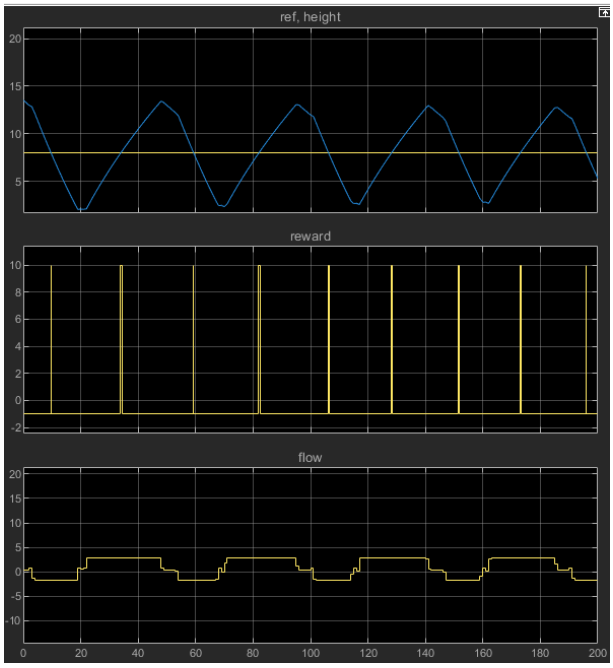
序号	学习参数数量(Actor/Critic)	结果特点	原因分析
1	1500/26	迭代次数增加 稳定性提高	见截图下的分析
2	1500/28	迭代次数增加 稳定性提高	见截图下的分析
3	1600/16	迭代次数减少 稳定性提高	见截图下的分析
4	1600/16	迭代次数增加 稳定性提高	见截图下的分析

1.3 作业2

本人希望调整窗口长度来达到使得平均奖励的波动变大或变小。

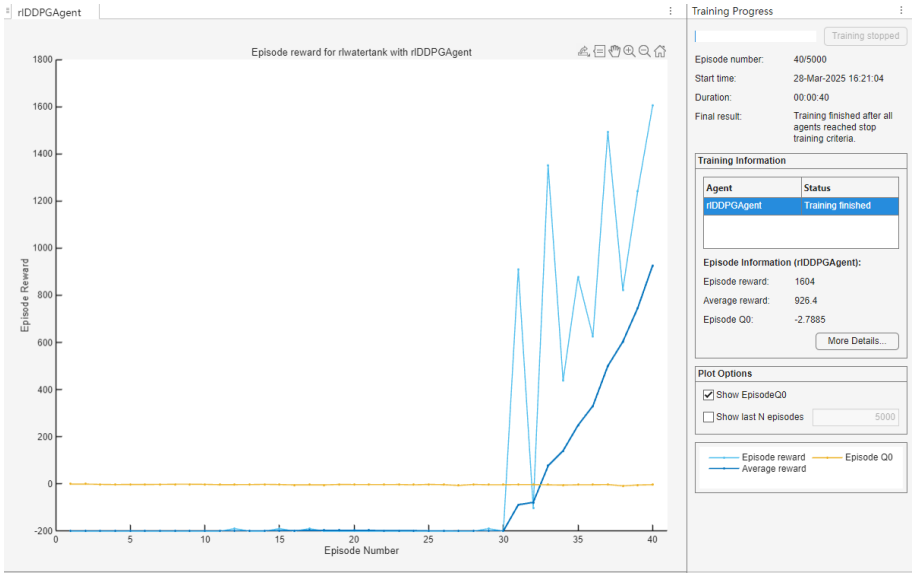
```
trainOpts = rlTrainingOptions(...
    MaxEpisodes=5000, ...
    MaxStepsPerEpisode=ceil(Tf/Ts), ...
    ScoreAveragingWindowLength=40, ...
    Verbose=false, ...
    Plots="training-progress",...
    StopTrainingCriteria="AverageReward",...
    StopTrainingValue=800);
```

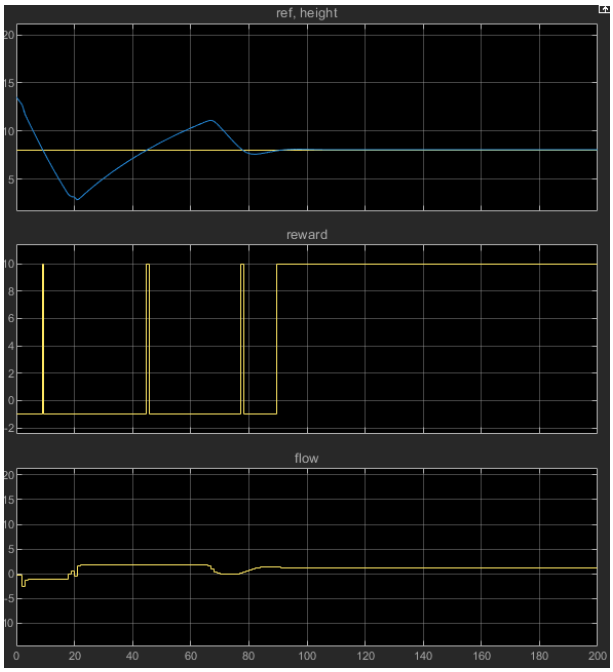




可以看到，当窗口长度增加为40时，迭代次数变多了（因为计算更复杂了），但是水位的波动也更加稳定了

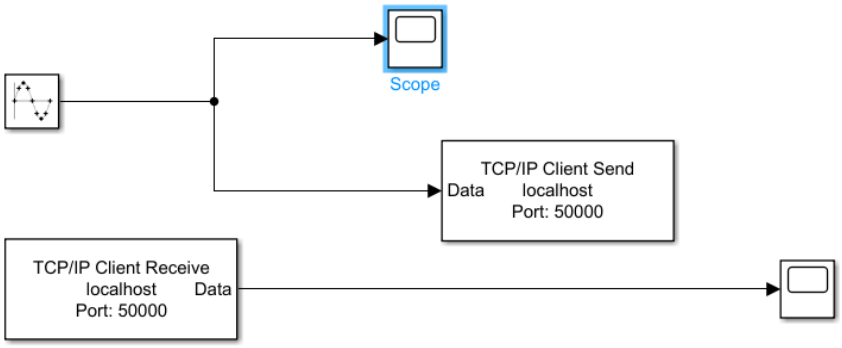
```
trainOpts = rlTrainingOptions(...
    MaxEpisodes=5000, ...
    MaxStepsPerEpisode=ceil(Tf/Ts), ...
    ScoreAveragingWindowLength=10, ...
    Verbose=false, ...
    Plots="training-progress",...
    StopTrainingCriteria="AverageReward",...
    StopTrainingValue=800);
```



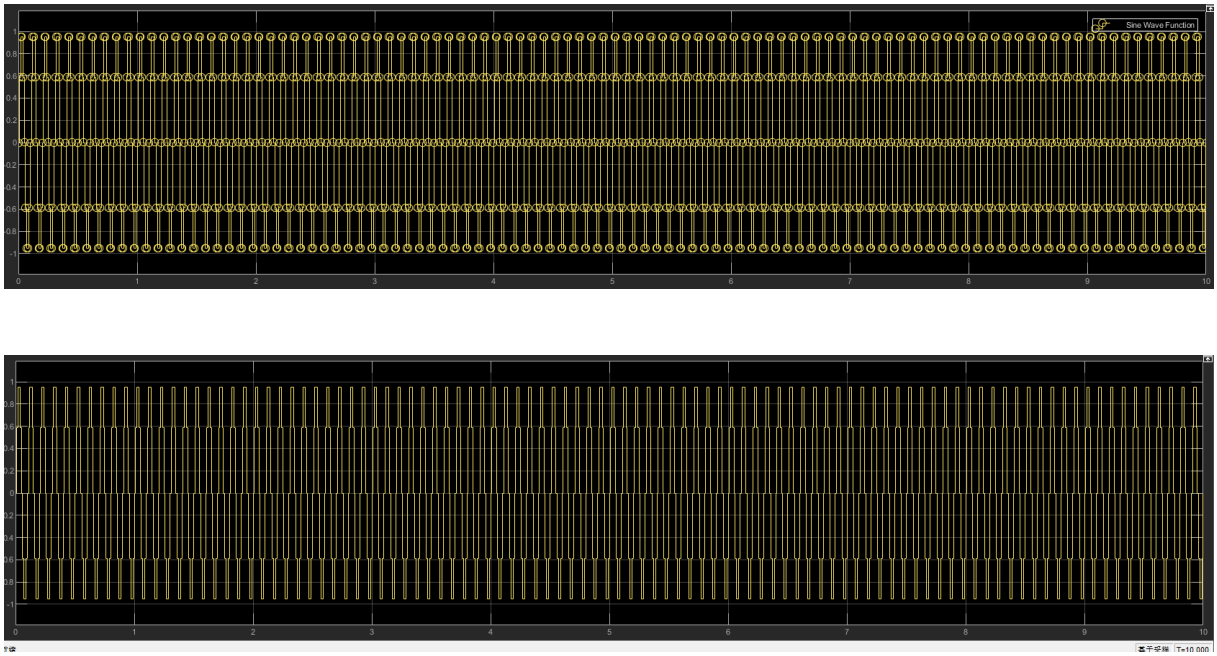


同理的，窗口长度降低到10时，迭代次数变少，同时水位波动也更加剧烈。

2 作业3



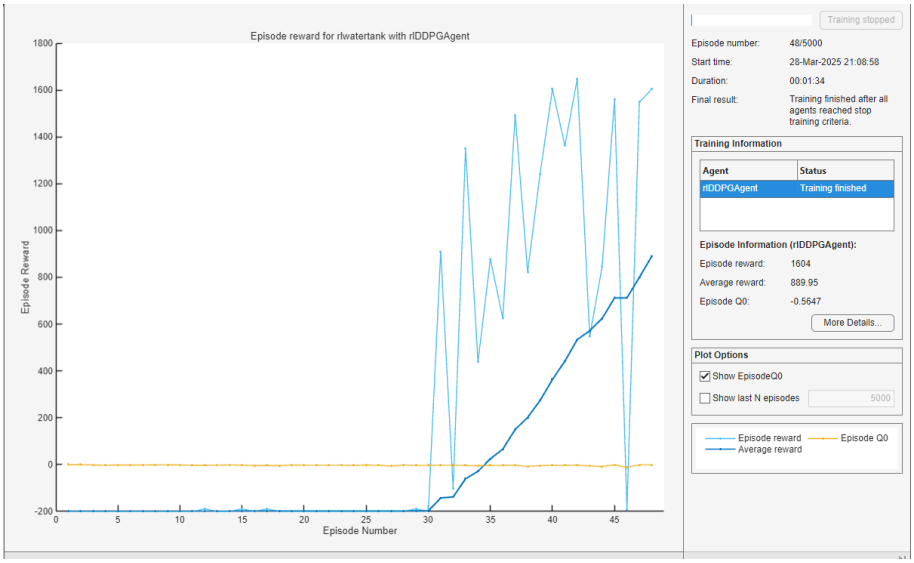
模型设计如图。



上图为发送和接受端的信号图，可以看到是一样的。

3 作业4

训练结果如下图:



	rlDDPGAgent
Status	Training finished
Episode number	48
Episode reward	1604
Episode steps	200
Total agent steps	9503
Average reward	889.95
Average steps	195.15
Episode Q0	-0.5647
Averaging window length	20
Training stopped by	AverageReward
Training stopped at	800

4 作业5

如图，自定义数据包含一个向量，一个常数以及一个布尔值。

